A scalable business framework for fungal mats as a sustainable alternative for topsoil preservation

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INTRODUCTION

- Topsoil erosion is an ongoing issue within agricultural areas.
- Eroded topsoil can lead to reduced agricultural production and a loss of food security (Zhang et al, 2021).
- Soil erosion can lead to habitat degradation and a loss of biodiversity.
- Topsoil erosion is mostly caused from rain, wind, and lack of vegetation (Zou et al, 2018).
- To address this issue, fungal mats can be installed in areas with high concentrations of topsoil loss.
- Fungal mats contain mycelium to bind soil particles together, acting as a thread (Fricker, et al., 2017).
- The web-like network prevents topsoil from being washed or blown away.
- Keeping topsoil in place will lead to an increased production of vegetation and crops.

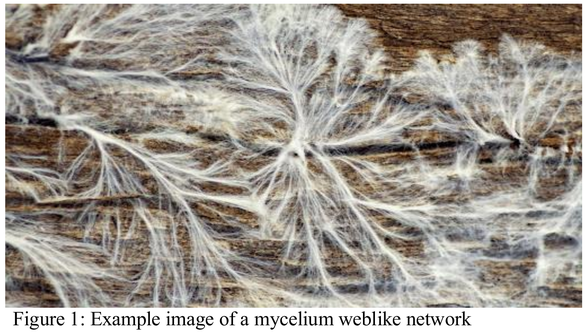

Figure 1: Example image of a mycelium weblike network

OBJECTIVES

- Create business model
- Estimating effects on erosion using fungal mats
- Model hydrological benefits, like increased water retention
- Estimating nutrient retention

METHODS

- To evaluate economic feasibility, a business model was created to estimate the cost of small and industrial scale scenarios using labor, materials, facilities/equipment, and overhead costs.
- Scaling-dependent efficiencies were considered due to automation of product and bulk procurement. Initial implementation costs was kept separate from per acre cost during calculations.
- A simplified Universal Soil Loss Equation approach was used to calculate the erosion and preserved value based on a national average erosion quantity, where average soil loss is equal to rainfall erosion rate, soil erodibility factor, slope length and steepness, cover crop management factor, and conservation support practice are all multiplied together. $Soil\ Loss\ (tons/acre/year) A = R \times K \times LS \times (C \times P)$
- Water retention (Pauwel et al., 2023; Salifu et al. 2022) and nutrient coefficients (Meena et al., 2017) were applied using published sources.
- Nationwide effects were found using the Industrial model. Per-acre annual benefit was scaled using assumed adopted acres. Adoption scenarios determined the range of fungal mat adoption and the overall benefit.
- The length of time each mat lasted was assumed at 5 years based on similar products. The 5-year lifetime was used to find lifetime benefit, lifetime cost, net benefit, and benefit-cost ratio.

$$A_{mats} = A_{baseline} \times C$$

$$\Delta A = A_{baseline} - A_{mats}$$

RESULTS

Table 1: Three scenarios highlighting the costs and benefits

Scenario	Soil preserved (tons/ac/yr)	Water (m3/ac/yr)	N (kg/ac/yr)	P (kg/ac/yr)	K (kg/ac/yr)
Conservative	3.54	166.94	3.54	1.416	0.708
Medium	3.95	166.94	3.95	1.58	0.79
Optimistic	4.36	166.94	4.36	1.744	0.872
	Total nutrients (kg/ac/yr)	Nutrient value ($)	Soil value ($)	Total benefit ($/yr)	Lifetime benefit (5 yr, $)
Conservative	5.664	10.69	184.08	194.77	973.85
Medium	6.32	11.93	205.4	217.33	1086.65
Optimistic	6.976	13.17	226.72	239.89	1199.45

Table 2: Small and industrial costs for different scenarios

Scenario	Labor	Substrate	Spawn	Netting
Small Scale	119	35	67.5	202.3
Industrial Scale	101.15	28	57.375	161.84
	Equipment	Total Cost	Payback (yrs)	Total Startup Cost (USD)
Small Scale	10	477.18	2.195647173	1869591.27
Industrial Scale	3	386.5015	1.778408411	11818324.27

Table 3: Costs and benefits of different rates of implementation

Adoption (%)	Acres treated (M)	Lifetime cost (B$)	Lifetime benefit (B$)	Net benefit (B$)	Benefit-Cost Ratio
5	19.5	7.55	21.19	13.64	2.81
10	39	15.09	42.37	27.28	2.81
25	97.5	37.73	105.95	68.22	2.81
50	195	75.46	211.89	136.43	2.81
100	390	150.93	423.78	272.85	2.81

CONCLUSION

- Fungal mats would preserve as substantial amount of soil on average (Table 1).
- Our findings indicate that water retention is increased due to the fungal networks of the mats (Table 1).
- Nutrient retention also increases with use of fungal mats (Table 1).
- It would not be out of reach to make this business based on our business model, due to billions of dollars being saved (Table 3)
- Our business model shows a relatively affordable start-up cost in comparison to the benefits of fungal mats (Table 2).

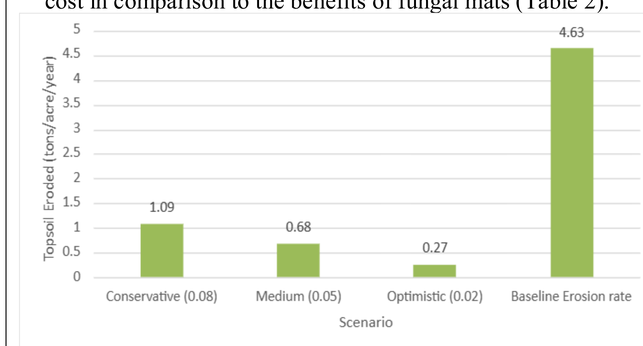

Figure 2: Topsoil erosion rates based off different scenarios and, in comparison to a baseline erosion rate

FUTURE WORK

- Further experimentation with fungal mats is needed to see their full effects and how they compare to other topsoil erosion mitigation strategies, like cover crops and mulching.

REFERENCES

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